

Module 5: Rise and fall of the clockwork universe

5.1 Models and rules

This module builds upon the work covered earlier in the course. The first section (5.1.1) uses simple techniques to model radioactive decay, capacitor charging and discharging and simple harmonic motion. In this framework, the formalism of the differential equation is developed along with the concept of field. There are many opportunities for practical work and empirical data can be compared and contrasted to the predictions made by the simple mathematical models. The field model is developed through consideration of gravitational fields.

The section raises questions about simplification in models, their usefulness and limitations (HSW7, HSW11). The difference can be highlighted between those models in which well-determined behaviour is due to exact rules operating on variables (as in

the harmonic oscillator) and those in which well-determined behaviour is due to smooth averages over many particles (as in radioactive decay).

The second section (5.1.2) develops ideas about gravitational field strength and potential. Space flight and astronomical data can provide a context and there are further opportunities to consider the development of the modern view of the universe (HSW1, HSW6, HSW7, HSW11).

The third section (5.1.3) covers a descriptive and mainly qualitative outline of the main features of the observable universe consistent with the hot big bang model of its origin. The ideas of the universality of the speed of light and the relativistic consequence of time dilation are introduced.

5.1.1 Creating models

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) capacitance as the ratio $C = \frac{Q}{V}$	
(ii) the energy on a capacitor $E = \frac{1}{2}QV$	HSW6
(iii) the exponential form of the decay of charge on a capacitor as due to the rate of removal of charge being proportional to the charge remaining $\frac{dQ}{dt} = -\frac{Q}{RC}$	exponential relationship as shown, explained using constant ratio property M3.9
(iv) the exponential form of radioactive decay as a random process with a fixed probability, the number of nuclei decaying being proportional to the number remaining $\frac{dN}{dt} = -\lambda N$	M1.3
(v) simple harmonic motion of a mass with a restoring force proportional to displacement such that $\frac{d^2x}{dt^2} = -kx$	

(vi) simple harmonic motion of a system where $a = -\omega^2 x$, where $\omega = 2\pi f$, and two possible solutions are $x = A \sin(\omega t)$ and $x = A \cos(\omega t)$

(vii) kinetic and potential energy changes in simple harmonic motion

example of conservation of energy
HSW1

(viii) free and forced vibrations, damping and resonance.

qualitative treatment only
HSW9

(b) *Make appropriate use of:*

(i) for a capacitor:
the term: time constant τ

(ii) for radioactive decay:
the terms: activity, decay constant λ , half-life $T_{\frac{1}{2}}$, probability, randomness

HSW8

(iii) for oscillating systems:
the terms: simple harmonic motion, period, frequency, free and forced oscillations, resonance, damping

HSW8

by expressing in words:

(iv) relationships of the form $\frac{dx}{dt} = -kx$, where rate of change is proportional to amount present

Learners are expected to be able to transfer relationships from words, formulae and diagrams, converting from any one form to another

by sketching, plotting from data and interpreting:

(v) exponential curves plotted with linear or logarithmic scales

Capacitor charging and discharging curves plotted against linear scales

M2.5, M3.10, M3.11, M3.12

(vi) energy of capacitor as area below a Q–V graph

M3.8
HSW5

(vii) x–t, v–t and a–t graphs of simple harmonic motion including their relative phases

M3.12
HSW5

(viii) amplitude of a resonator against driving frequency.

(c) *Make calculations and estimates involving:*

(i) calculating activity and half-life of a radioactive source from data,

HSW10

$$T_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

(ii) solving equations of the form $\frac{\Delta N}{\Delta t} = -\lambda N$ by iterative numerical or graphical methods	$N = N_0 e^{-\lambda t}$ as the analytic solution M3.9 HSW3
(iii) calculating time constant τ of a capacitor circuit from data; $\tau = RC$; $Q = Q_0 e^{-t/RC}$	M0.5
(iv) solving equations of the form $\frac{\Delta Q}{\Delta t} = -\frac{Q}{RC}$ discharging $Q = Q_0 e^{-t/RC}$; charging $Q = Q_0(1 - e^{-t/RC})$; corresponding equations for V and I	by iterative numerical/graphical methods M0.5, M3.9
(v) $C = \frac{Q}{V}$, $I = \frac{\Delta Q}{\Delta t}$, $E = \frac{1}{2}QV = \frac{1}{2}CV^2$	M2.3
(vi) $T = 2\pi\sqrt{\frac{m}{k}}$ with $f = 1/T$ for a mass oscillating on a spring	M2.2
(vii) $T = 2\pi\sqrt{\frac{L}{g}}$ for a simple pendulum	M2.2
(viii) $F = kx$; $E = \frac{1}{2}kx^2$	
(ix) solving equations of the form $\frac{\Delta^2 x}{\Delta t^2} = -\frac{k}{m}x$ by iterative numerical or graphical methods	M3.9
(x) $x = A \sin 2\pi ft$ or $x = A \cos 2\pi ft$	M0.6, M4.5
(xi) $E_{\text{total}} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$.	
(d) <i>Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):</i>	
(i) measuring the period/frequency of simple harmonic oscillations for example mass on a spring or simple pendulum and relating this to parameters such as mass and length	links to 5.1.1a(v), (vi), (vii), b(iii), (iv), c(vi), (vii), (viii), (ix), (x), (xi) PAG10
(ii) qualitative observations of forced and damped oscillations for a range of systems	links to 5.1.1a(viii), b(iii) PAG10
(iii) investigating the charging and discharging of a capacitor using both meters and data loggers	links to 5.1.1a(iii), b(v), c(iii), c(iv) PAG9
(iv) determining the half-life of an isotope such as protactinium.	links to 5.1.1c(i) PAG7